    - These methods learn the optimal policy indirectly by maximizing value functions.
2. **Policy-Based Methods**:
    - Learn the policy directly without requiring value functions.
    - Examples include Policy Gradient methods, which optimize policies through gradient ascent.
3. **Model-Based Methods**:
    - Build a model of the environment's dynamics and use it to plan actions. These methods can be computationally intensive but offer efficiency in certain tasks.

Applications of Reinforcement Learning

Reinforcement Learning has a wide range of real-world applications, including:

1. **Game Playing**:
    - RL has achieved superhuman performance in games like Chess, Go, and StarCraft.
2. **Robotics**:
    - Teaching robots to walk, grasp objects, or navigate obstacles.
3. **Healthcare**:
    - Optimizing treatment plans and resource allocation in hospitals.
4. **Finance**:
    - Developing trading strategies and portfolio optimization.
5. **Autonomous Vehicles**:
    - Enabling self-driving cars to make real-time decisions in complex traffic scenarios.